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The influence of silicon and aluminum alloying on the lattice parameter and stacking fault energy of austenitic steel

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The influence of Si and Al on the lattice parameter and stacking fault energy (SFE) of austenitic steel was studied. X-ray diffraction indicated that 2.5 wt.% Al alloying increased the austenite lattice parameter by 1.41×10^{-3} nm, whereas 2.5 wt.% Si alloying had little effect. Partial dislocation separation measurements from weak-beam dark-field images indicated that 2.5 wt.% Al alloying increased the SFE by 12–17 mJ m^{-2} and 2.5 wt.% Si alloying decreased the SFE by 6–7 mJ m^{-2}.
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Advanced high-strength steels (AHSS) containing metastable austenite are candidate materials for automobile weight reduction [1–8]. Recent investigations regarding TRIP steels have evaluated the substitution of Al for Si alloying. Both Si and Al inhibit carbide formation during austempering and promote austenite retention, but Al alloying accelerates bainitic transformation during austempering [9,10] and results in better coatability [11,12]. Al alloying in TRIP steels also increases resistance to strain-induced α′ martensitic transformation relative to Si-alloyed grades [13,14] possibly due to increases in carbon partitioning to austenite during austempering [14,15] and increases in austenite stacking fault energy (SFE) [13]. This paper focuses on the effects of Al and Si alloying on the measured austenite carbon content and SFE.

Carbon is a potent austenite stabilizer, and its concentration in retained austenite is commonly ascertained by X-ray diffraction (XRD) measurements. Linear correlations between austenite lattice parameter (a_o) and carbon content are proposed in the literature [16–19]. However, Al is also reported to expand the austenite lattice [16], but its influence is sometimes excluded when calculating the austenite C content. Thus, the austenite C content in Al-alloyed AHSS may be overestimated, which steels, and thus may be misleading with regard to the effect of Al on carbon partitioning to austenite. In contrast, Si is reported to have a negligible influence on the austenite lattice parameter [16].

The SFE also affects austenite stability through its effect on shear band formation [20]. Al alloying may stabilize austenite by increasing the austenite SFE [21–24]. Experimental transmission electron microscopy (TEM) [25] and XRD [26–28] studies indicate that Si decreases the SFE of austenite, while thermochemical modeling indicates that Si decreases the SFE slightly [29] or that the influence of Si on the SFE is non-monotonic [21].

In this study, the influences of Si and Al on the austenite lattice parameter and SFE are investigated using the three austenitic steel compositions shown in Table 1, which were derived from a microstructure survey of the Fe–Ni–Cr–Al–Si system [30]. The compositions are all approximately the same except for variations in Al and Si alloying. The three steels were vacuum cast, homogenized for 20 h at 1200 °C, hot rolled and then annealed for 1 h at the temperatures indicated in Table 1 to achieve coarse grain sizes of approximately 140 μm.

Fully austenitic steels were employed because no assumptions regarding element partitioning in a multiphase microstructure need to be made in the XRD analysis. Furthermore, TEM analysis of dislocations for SFE determination is performed more easily with a coarse-grained single-phase microstructure. However, the assumptions and possible limitations of extending the analysis to multiphase AHSS must be acknowledged. Austenite is retained to room temperature in the experimental alloys by Ni and Cr alloying, not by carbon partitioning, as in AHSS. Therefore, to extend the results of this study to low-alloy AHSS, the effects of Si and Al on austenite characteristics are assumed to be independent of Ni and Cr alloying.

Seven coupons for XRD examination were taken from various locations in each alloy. The coupons were ground

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Table 1. Experimental steel compositions (wt.%) and solution annealing temperatures.

Alloy	C	N	Ni	Cr	Mn	Si	Al	Annealing Temp. (°C)
Base	0.028	0.0034	15.05	11.06	1.09	0.002	0.05	1040
2.5 Si	0.028	0.0036	14.99	10.98	1.10	2.50	0.06	1075
2.5 Al	0.030	0.0034	15.04	11.02	1.08	0.034	2.47	1060

to a 600 grit surface finish and then chemically polished for 20 min in a solution of 10 parts water, 10 parts hydrogen peroxide and 1 part hydrofluoric acid. A Phillips X'Pert® diffractometer with Cu K_α ($\lambda = 0.154060$ nm) radiation was used for the XRD. HighScore Plus® 3.0.5 software was used for analysis. The lattice parameters (a_0) determined from the first four austenite peaks ((111), (200), (220) and (311)) were plotted against $\cos^2\theta/\sin\theta$, and the a_0 for each specimen was determined from a linear regression extrapolation to $\theta = 90°$ [31]. The a_0 vs. $\cos^2\theta/\sin\theta$ regression analysis was weighted to favor higher angle peaks by using quadruplicate data points for (311) peaks, triplicate data points for (220) peaks, duplicate data points for (200) peaks and single data points for (111) peaks [32]. The lattice parameter for each material was then obtained by discarding the lowest and highest values of a_0, and averaging the results of the remaining five specimens.

For the SFE determination, 12.7 mm diameter cylinders of each material were machined with the cylinder axes parallel to the rolling direction. The cylinders were compressed to 5% true strain to introduce dislocations into the specimens. Thin slices were then sectioned from the cylinders at 45° from the compression axis and prepared for TEM by grinding and electropolishing with 95% acetic acid and 5% perchloric acid.

TEM investigation was performed in a Phillips CM12® TEM operating at 120 kV. For each material, grains with {111} planes nearly aligned with the plane of the foil were used for weak-beam dark-field (WBDF) imaging of dissociated dislocations [33,34]. The WBDF imaging was performed using **g**3**g** conditions with (220)-type reflections ($s_g \approx 0.22\ \text{nm}^{-1}$). Grains near a {111} orientation were selected because they had favorable slip conditions during compression and would therefore contain dislocations. Furthermore, dissociated dislocations imaged in this configuration are minimally inclined to the electron beam. Thus, the measured partial dislocation separations do not need to be corrected for 2-D image projection. The evaluated dislocations were confirmed to exist in the {111} plane approximately normal to the electron beam through $\mathbf{g}.\mathbf{b_p}$ analysis of the partial dislocations [33,34].

For each material, multiple data pairs of partial dislocation separation, d, and perfect dislocation character angle, α, were recorded along several dislocations. Figure 1 shows an example WBDF image of a dissociated dislocation in the base alloy, along with representations of d and α. As shown in Figure 1, α is the angle between the line vector and the Burgers vector, **b**, of the undissociated dislocation. Because both partial dislocations are in contrast in the condition shown and because $\mathbf{g}.\mathbf{b_p}$ analysis was performed, it is known that the **g** vector used to form the two-beam condition is parallel to the undissociated dislocation Burgers vector, **b**. The paired (d, α) data were then input into Eq. (1) [35] to calculate the SFE values for each measured data pair:

$$SFE = \frac{G|a_0|^2(2-\nu)}{48\pi(1-\nu)} \cdot \frac{1}{d} \cdot \left(1 - \frac{2\nu\cos 2\alpha}{2-\nu}\right) \quad (1)$$

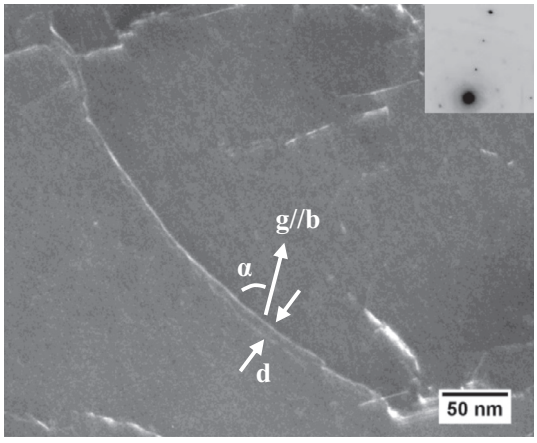


Figure 1. WBDF TEM micrograph of a dissociated dislocation in the base alloy. The annotations are explained in the text.

These values were then averaged to determine the SFE for each material. The experimental lattice parameters (a_0) determined from the XRD analysis were used in Eq. (1). The isotropic shear modulus (G) and Poisson ratio (ν) were assumed to be the same for each material, and were 74 GPa and 0.3, respectively [36]. In order to estimate the possible effect of elastic anisotropy on {111} dislocations, the effective G (66 GPa) and ν (0.43) values determined for a 12 Ni–12 Cr steel [37,38] were also applied to the experimental WBDF data, which increased the calculated SFE value for the 2.5 Al alloy by approximately 10%, but had little effect on the other two alloys. Since elastic anisotropy was not evaluated in this study and since other literature reports an effective ν closer to 0.3 instead of 0.43 [29], the isotropic elastic properties were used in the SFE calculations.

Figure 2 and Table 2 show the calculated austenite lattice parameter as a function of relative Si and Al alloying. Relative Si alloying refers to the difference in Si content between the 2.5 Si and base alloys, and the relative Al alloying refers to the difference in Al content between the two alloys. Because all of the alloys have approximately

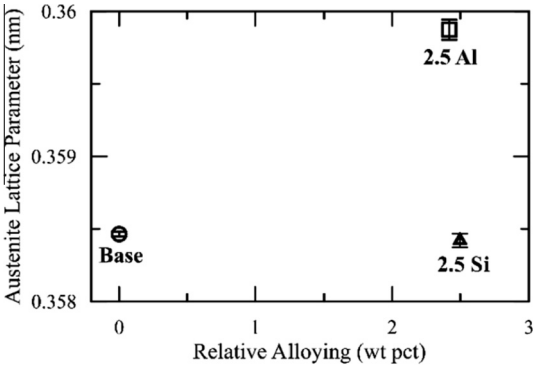


Figure 2. The dependence of austenite lattice parameter on Si and Al alloying.

Table 2. Empirical Si and Al linear elemental coefficients for austenite lattice parameter and SFE.

Element	A^{LP} (10^{-3} nm wt.%$^{-1}$) [10^{-3} nm at.%$^{-1}$]	A^{LP} (10^{-3} nm wt.%$^{-1}$) [10^{-3} nm at.%$^{-1}$] From Dyson and Holmes [16]	A^{SFE} (mJ m^{-2} wt.%$^{-1}$) [10^{-3} nm at.%$^{-1}$]
Si	−0.018 [−0.009]	≈0 [≈0]	−2.6 [−1.3]
Al	0.582 [0.289]	0.56 [0.28]	4.9 [2.5]

equal contents of all elements other than Si and Al (Table 1), only differences in Si and Al content have been considered. Figure 2 indicates that 2.5 wt.% Al increases the austenite lattice parameter by 1.41×10^{-3} nm relative to the base alloy, while 2.5 wt.% Si has a minimal effect. Linear empirical relations between the austenite composition and the lattice parameter for austenitic steels are reported in the literature [16] and take the form $a_0 = \sum_i A_i^{LP} x_i$, where x_i is the wt.% of element i in austenite and A_i^{LP} is the linear lattice parameter coefficient for element i, in units of 10^{-3} nm wt.%$^{-1}$. The calculated A^{LP} for Si and Al are −0.018 and 0.582 10^{-3} nm wt.%$^{-1}$, respectively. These values agree well with those reported by Dyson and Holmes [16] (Table 2).

Since Al alloying increases the austenite lattice parameter, it is important to account for Al in calculations of the austenite C content based on the lattice parameter. If only carbon is considered to expand the lattice, the calculated C content may be an overestimate. Values for A_C^{LP} reported in the literature range from 3.30×10^{-3} nm wt.%$^{-1}$ for austenitic stainless steels to 4.67×10^{-3} nm wt.%$^{-1}$ for low-alloy steels [16–19]. If Al alloying is ignored, the overestimate of austenite C content, δ_C, can be expressed as $\delta_C = x_{Al}(A_{Al}^{LP}/A_C^{LP})$. Al-alloying in TRIP steels is often on the order of 1 wt.%. Using the value of A_{Al}^{LP} found in this study, the range of A_C^{LP} values reported in the literature, and assuming x_{Al} is 1 wt.%, the value of δ_C ranges from 0.125 to 0.176 wt.%.

The magnitude of δ_C is non-negligible. An increase in retained austenite carbon content in multiphase TRIP steels from 0.09 to 0.21 wt.% has been shown to increase its thermodynamic stability [39]. This carbon content range is on the same order of magnitude as the δ_C estimations above. Thus, the influence of Al on the measured retained austenite lattice parameter should not be ignored. It should also be noted that the actual value of δ_C for AHSS depends on the nominal steel composition and thermal processing schedule. For example, in intercritically annealed TRIP steel, Al may partition to the intercritical ferrite, resulting in a retained austenite Al content, x_{Al}, that is lower than the nominal steel Al content. Conversely, if the steel is instead fully austenitized before lower temperature carbon partitioning, the Al content of the austenite retained to room temperature should equal the nominal steel composition because limited substitutional element partitioning is expected during such carbon partitioning [40].

Composition also influences austenite stability through its effect on the thermodynamics of the austenite-to-epsilon martensite transformation and the SFE. Lower austenite SFE values promote the formation of planar shear bands, including epsilon martensite, the intersections of which may serve as α′ martensite nuclei [20]. Increasing the austenite SFE is expected to suppress shear band formation and strain-induced martensite formation. Therefore, it is important to characterize the effects of Al and Si alloying on the SFE.

Figure 3(a) shows the measurements of partial dislocation spacing as a function of dislocation angle for the base, 2.5 Si and 2.5 Al alloys. In Figure 3(a), the solid lines represent the average calculated SFE for each material and the dashed lines indicate the upper and lower bounds from the calculated SFE standard deviations. The base alloy SFE is 26.3 mJ m^{-2}, which is consistent with the value of 30 mJ m^{-2} found for a similar 15 Ni–11 Cr alloy through TEM analysis of extended dislocation nodes [25]. Figure 3(b) shows the dependence of SFE on relative Si and Al alloying. Figure 3(b) indicates that 2.5 wt.% Al alloying increases the SFE by 12.0 mJ m^{-2} from the base alloy value whereas 2.5 wt.% Si alloying decreases it by 6.4 mJ m^{-2}. A correction is available to account for the effect of the partial dislocation strain fields on the observed partial dislocation

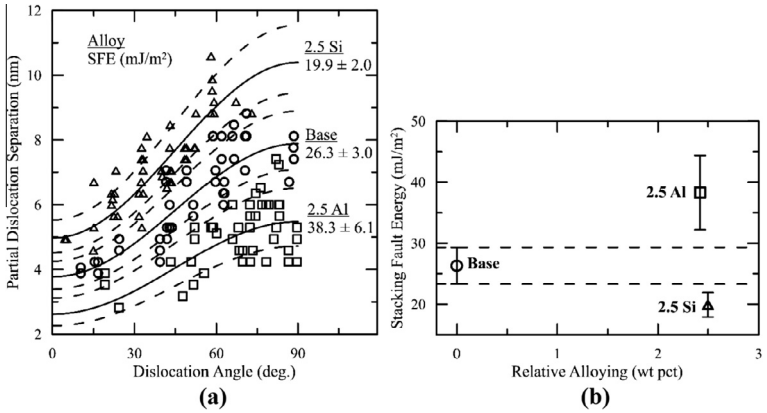


Figure 3. (a) Dependence of partial dislocation separation on perfect dislocation character angle for the base, 2.5 Si and 2.5 Al alloys. The symbols indicate the measured data while the solid lines indicate the calculated trends based on the calculated average SFE for each material. The dashed lines indicate the upper and lower bounds determined from the calculated SFE standard deviation for each material. (b) Dependence of SFE on Si and Al alloying.

spacings and thus the measured SFE values [34]. The correction suggested by Williams and Carter [34] was not applied because not all of the TEM data was available to assign all of the necessary parameters. An upper limit estimation suggests that the biggest change is in the 2.5 Al alloy, where the correction would increase the SFE of the 2.5 Al alloy to 44.0 ± 9.8 mJ m^{-2}. The SFEs of the other alloys would change by approximately 1 mJ m^{-2} or less. Since the corrected values are estimated, the uncorrected data will be used for the remainder of the paper unless otherwise noted.

Al has also been reported in the literature to increase the SFE [21–24], which is consistent with the current observations. However, there are conflicting reports of the effect of Si on austenite SFE; some sources indicate that Si decreases the SFE [25–29], which is supported by the current study, while others suggest that the dependence of the SFE on Si alloying is non-monotonic [21]. Linear empirical relationships between austenite composition and SFE have been proposed [25,26] and take the form of $\mathrm{SFE} = \sum_i A_i^{SFE} x_i$, where x_i is the wt.% of element i in austenite and A_i^{SFE} is the linear SFE coefficient for element i, in units of mJ m^{-2} wt.%$^{-1}$. Table 2 contains the values of A_{Si}^{SFE} and A_{Al}^{SFE} found in this study.

The values of A_{Si}^{SFE} reported in the high-Mn steel literature vary from -1 to -3.5 mJ m^{-2} wt.%$^{-1}$ [27–29], which is in good agreement with the value of -2.6 mJ m^{-2} wt.%$^{-1}$ found in this study (Table 2). Thermochemical modeling of the SFE in high-Mn alloys predicts an A_{Al}^{SFE} value of approximately 5.2 mJ m^{-2} wt.%$^{-1}$ [21], which is in very good agreement with the value of 4.9 mJ m^{-2} wt.%$^{-1}$ found in this study (Table 2). Jeong et al. [28] found an A_{Al}^{SFE} value of approximately 6.4 mJ m^{-2} wt.%$^{-1}$ in high-Mn alloys using XRD analysis. Considering the influence of the partial dislocation strain fields on their apparent position (and the calculated SFE) for the base and 2.5 Al alloys yields an estimated A_{Al}^{SFE} value of 6.9 mJ m^{-2} wt.%$^{-1}$ in this study. In contrast, Kim et al. [23] showed, using TEM WBDF imaging of dissociated dislocations, that Al increases the SFE in high-Mn steels by 11.3 mJ m^{-2} wt.%$^{-1}$, which is higher than the values found in this study.

In conclusion, it was found that 2.5 wt.% Al alloying increases the austenite lattice parameter by 1.41×10^{-3} nm whereas Si has little effect. Furthermore, ignoring the effects of Al alloying on the austenite lattice parameter may result in an overestimation of the C content on the order of 0.15 wt.% in Al-alloyed AHSS. Finally, 2.5 wt.% Al alloying increases SFE by 12–17 mJ m^{-2} and 2.5 wt.% Si alloying decreases it by 6–7 mJ m^{-2} in the alloy system investigated. These observations are generally supported by other reports of alloying effects on SFE.

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